Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

Вахиш Дуаа Иссам Али Ахмед

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
duaavahish@duaavahish:~$ ls /etc/ > file.txt
duaavahish@duaavahish:~$ ls >> file.txt
duaavahish@duaavahish:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
duaavahish@duaavahish:~$ grep .conf file.txt > conf.txt
duaavahish@duaavahish:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/duaavahish/.npm/_cacache/index-v5/09/c0
/home/duaavahish/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ce
d5cf6fc29def30220242c53
/home/duaavahish/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a589
96cfad55fb2000f59e2edcc
/home/duaavahish/.npm/_cacache/index-v5/cd
/home/duaavahish/.npm/_cacache/index-v5/c9
/home/duaavahish/.npm/_cacache/index-v5/c0
/home/duaavahish/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b
53cea310cda0b203217f693
/home/duaavahish/.npm/_cacache/index-v5/af/cf
/home/duaavahish/.npm/_cacache/index-v5/af/3b/c01fad9ea9ac6256e90d5a59e2c8b816cc7bc
f0f41a5ca46eaa0bbab8743
/home/duaavahish/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86
117d1350482a1d990504757
/home/duaavahish/.npm/_cacache/index-v5/c3
/home/duaavahish/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca8753242
2f2506214f94db8861160d3
/home/duaavahish/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837
bc2c8188a44d554f3f3ba2e
/home/duaavahish/.npm/_cacache/index-v5/ca
/home/duaavahish/.npm/_cacache/index-v5/95/f7/cabf02718d2ced903e42baa9d89ccac492bdd
331ae80e550538ac79b0d3a
/home/duaavahish/snap/hugo/common
/home/duaavahish/snap/hugo/current
/home/duaavahish/conf.txt
duaavahish@duaavahish:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
:
```

Рисунок 4: Поиск файлов

Управление процессами

```
duaavahish@duaavahish:~$ dind ~ -name "log*" > logfile &
[1] 6897
bash: dind: command not found...
duaavahish@duaavahish:~$ find ~ -name "log*" > logfile &
[2] 6924
[1]   Exit 127                dind ~ -name "log*" > logfile
duaavahish@duaavahish:~$
[2]+  Done                    find ~ -name "log*" > logfile
duaavahish@duaavahish:~$ rm logfile
duaavahish@duaavahish:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
duaavahish@duaavahish:~$
duaavahish@duaavahish:~$ gedit &
[1] 6962
duaavahish@duaavahish:~$ ps | grep gedit
   6962 pts/1    00:00:00 gedit
duaavahish@duaavahish:~$ kill 6962
[1]+  Terminated              gedit
duaavahish@duaavahish:~$
duaavahish@duaavahish:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page documents the GNU version of df.  df displays the amount
       of space available on the file system containing each file  name  argument.
       If no file name is given, the space available on all currently mounted file
       systems is shown.  Space is shown in 1K blocks by default, unless the envi-
       ronment  variable POSIXLY_CORRECT is set, in which case 512-byte blocks are
       used.

       If an argument is the absolute file name of  a  device  node  containing  a
       mounted  file  system,  df  shows  the  space available on that file system
       rather than on the file system containing the device node.  This version of
       df cannot show the space available on unmounted file  systems,  because  on
       most  kinds of systems doing so requires non-portable intimate knowledge of
       file system structures.

OPTIONS
       Show information about the file system on which each FILE resides,  or  all
       file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes
              in units of 1,048,576 bytes; see SIZE format below
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                          User Commands                          DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print apparent sizes rather than device usage; although the apparent
              size is usually smaller, it may be larger due to holes in ('sparse')
              files, internal fragmentation, indirect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by SIZE before printing them; e.g., '-BM' prints sizes
              in units of 1,048,576 bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
              produce a grand total

 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
duaavahish@duaavahish:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89485636  65028908  58% /
devtmpfs         4011276        0   4011276   0% /dev
tmpfs            4035652       92   4035560   1% /dev/shm
tmpfs            1614264     1948   1612316   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.
service
tmpfs            4035652     1640   4034012   1% /tmp
/dev/nvme0n1p3 155186176 89485636  65028908  58% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop1         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.
service
tmpfs             807128      168    806960   1% /run/user/1024
/dev/sr0         2677920  2677920         0 100% /run/media/duaavahish/Fedora-WS-Li
ve-43
duaavahish@duaavahish:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4        ./.npm/_cacache/index-v5/30/1a
4        ./.npm/_cacache/index-v5/30
424      ./.npm/_cacache/index-v5
115396   ./.npm/_cacache
115460   ./.npm
16       ./snap/hugo/25676/.config/go/telemetry/local
0        ./snap/hugo/25676/.config/go/telemetry/upload
16       ./snap/hugo/25676/.config/go/telemetry
16       ./snap/hugo/25676/.config/go
16       ./snap/hugo/25676/.config
16       ./snap/hugo/25676
0        ./snap/hugo/common
20       ./snap/hugo
20       ./snap
0        ./monthly
0        ./reports/monthly/monthly
0        ./reports/monthly
0        ./reports
4        ./ski.plases/equipment
0        ./ski.plases/plans
4        ./ski.plases
0        ./australia
0        ./play/games/play
0        ./play/games
0        ./play
1524448  .
duaavahish@duaavahish:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/duaavahish/.npm/_cacache/index-v5/9c/e7
/home/duaavahish/.npm/_cacache/index-v5/30
/home/duaavahish/.npm/_cacache/index-v5/30/1a
/home/duaavahish/snap
/home/duaavahish/snap/hugo
/home/duaavahish/snap/hugo/25676
/home/duaavahish/snap/hugo/25676/.config
/home/duaavahish/snap/hugo/25676/.config/go
/home/duaavahish/snap/hugo/25676/.config/go/telemetry
/home/duaavahish/snap/hugo/25676/.config/go/telemetry/local
/home/duaavahish/snap/hugo/25676/.config/go/telemetry/upload
/home/duaavahish/snap/hugo/common
/home/duaavahish/monthly
/home/duaavahish/reports
/home/duaavahish/reports/monthly
/home/duaavahish/reports/monthly/monthly
/home/duaavahish/ski.plases
/home/duaavahish/ski.plases/equipment
/home/duaavahish/ski.plases/plans
/home/duaavahish/australia
/home/duaavahish/play
/home/duaavahish/play/games
/home/duaavahish/play/games/play
duaavahish@duaavahish:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.